

Jenny Petrova

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Education

- PhD in Financial Mathematics, Florida State University** Aug 2024 – May 2029
- Current research: Neural methods for rare-event derivative pricing: deep parameterization of importance-sampling tilts, gradient-based training on variance objectives, and the stochastic approximation theory behind optimizer stability.
- MS in Financial Engineering, Florida State University** Aug 2024 – May 2026
GPA: 3.9/4.0
- Coursework: Numerical Analysis, Advanced Probability & Inference, Stochastic Optimal Control, Measure & Integration, Numerical Optimization, Stochastic Optimization, Monte Carlo Methods, Stochastic Analysis, Deep & Reinforcement Learning
- BS in Applied & Computational Mathematics, Florida State University** Aug 2020 – May 2024
Minor in Interdisciplinary Sciences GPA: 3.8/4.0
- Coursework: Mathematical Modeling, Linear Algebra, Partial Differential Equations, Complex Variables, Real Analysis
- BS in Statistics, Florida State University** Aug 2020 – May 2024
Minor in Business Analytics GPA: 3.6/4.0
- Coursework: Mathematical Statistics, Statistical Modeling, Statistics for Data Analytics, Data Science with Python

Experience

- Charles Schwab** Denver, CO
Quantitative Risk Analyst Intern Jun 2026 – Aug 2026
- Built digital option pricing framework for new product launch, implementing reliable pricing and sensitivity computation strategies by time to expiry. Constructed model validation tests and wrote documentation supporting the model risk oversight review.
 - Designed benchmark suite evaluating internal AI models across standardized competency and performance metrics.
 - 1st Place in Schwab Intern AI Competition, used internal AI API to integrate LLM agent into customer service platform.
- Florida State University** Tallahassee, FL
Graduate Teaching Assistant Aug 2024 – May 2026
- Served as sole instructor of record for Calculus I, teaching classes of ~30 students with full course ownership.
 - Developed all lecture notes, quizzes, midterms, and final exams, and managed complete course grading.
- Responsible AI Lab @ FSU** Tallahassee, FL
Research Assistant Dec 2025 – May 2026
- Contributed to a PyTorch-based Python library for natural hazard prediction.
- Cboe Global Markets** Chicago, IL
Global Derivatives Analyst — US Futures Intern May 2025 – Aug 2025
- Analyzed relationships between front-month futures settlement prices and underlying index movements to identify pricing patterns and potential inefficiencies.
 - Aggregated Lead Market Maker data to support initiatives enhancing proprietary product liquidity.
 - 1st Place in Cboe Intern AI Competition, developing a prototype AI-driven tool and demonstrating its potential to drive revenue growth.
- StraightForward Concepts** Chicago, IL
Data Science Analyst Intern May 2023 – May 2024
- Constructed psychological and behavioral segmentation profiles tailored to client needs.
 - Applied k -means clustering in Python to survey data to identify optimal market segments, then incorporated cluster labels into an XGBoost model to improve segmentation accuracy and predictive performance.

Research

- Deep Learning Optimization for Pricing Path-Dependent Options via Importance Sampling**
- Developed a Python framework that learns the optimal tilting parameter under a twisted measure, using SGD/Adam to minimize the second moment of the importance-sampling estimator and improve Monte Carlo efficiency for pricing rare-event, path-dependent options.
 - Found that heuristic initialization is load-bearing under high gradient noise, particularly with single-path gradient estimates.

Fast Graph-Based Algorithms for Image Segmentation

- Developed a Python algorithm mapping digital images to weighted graphs stored as matrices, demonstrating matrix-transformation methods for image segmentation.
- Presented at Troy University’s “MathFest” and Florida State University’s “Undergraduate Math Symposium.”

Modularity-Based Algorithms for Community Detection

- Studied spectral optimization methods and the Louvain algorithm for community detection, highlighting their effectiveness in image segmentation.

Equities Portfolio Optimization using James-Stein Shrinkage

- Constructed GMV and maximum-Sharpe-ratio portfolios across multiple time series, applying James-Stein shrinkage to improve return estimates for S&P 500 allocation.

Certifications

<i>Akuna Capital Options 201</i>	Sep 2025
<i>SAS Programming and Data Analysis</i>	May 2022

Skills

Programming: Python, Julia, C++, SQL, SAS
Computer: PyCharm, VS Code, Jupyter Notebook, AWS, GitHub, Microsoft Office, Tableau, Sigma
Languages: Bulgarian (<i>Bilingual</i>), Spanish (<i>Professional Working Proficiency</i>), Russian (<i>Conversational</i>)